EDUHUB / ENGINEERING DOCUMENTATION

TESTING & VALIDATION

EduHub | Online Learning Platform

83	32	48
FUNCTIONAL REQUIREMENTS	QUALITY REQUIREMENTS	MAPPED USE CASES

Candidate test strategy, runnable test-case design, risk-based validation, quality gates, user acceptance testing and traceability. No software execution or passing results are claimed.

BASELINE / ORIGINAL 5-PAGE PROJECT BRIEF + EDUHUB SRS / USE-CASE MODEL

Design status: proposed verification plan. Team approval is required for numeric quality gates.

SYSTEM ANALYSIS & DESIGN / NOV 2026

Document identity & status

Formal candidate test plan and validation baseline

83
SRS FRS

32
SRS NFRS

48
USE CASES

DOCUMENT CONTROL

ATTRIBUTE	SPECIFICATION
Document ID	EDH-SAD-TV-001 / Testing & Validation
Version	1.0 / Candidate baseline - review required
Phase	System Analysis & Design / Additional Deliverables
Primary source	EduHub project brief, 5 pages; source features/roles/integrations
Related artifacts	Requirements SRS (FR-001..083, NFR-01..32), Use Cases (UC-01..48), architecture, schema and UML behavior diagrams
Target solution	ASP.NET Core MVC / EduHub.Web + EduHub.Services + EduHub.Data / EF Core + SQL Server / Stripe / Azure
Audience	Development team, QA owner, supervisor, Student/Instructor/Admin UAT representatives
Execution status	NOT RUN. This PDF defines procedures and acceptance gates; it is not a test-results report.

Evidence policy BRIEF = explicitly stated in the user project file. SRS = previously proposed measurable requirement. PLAN = test design or tooling recommendation. Numeric targets are candidates until accepted by the team.

Testing objective & boundary

Verify EduHub behavior, operational safety, and readiness for stakeholder acceptance

WHY THIS DELIVERABLE EXISTS

EduHub must support trusted course publishing, free/paid enrollments, secure video access, progress and quiz grading, and verifiable certificates. Validation must provide repeatable evidence that the system behaves correctly across the Student, Instructor, Admin roles and its external providers. [S1, S2]

VERIFICATION VERSUS VALIDATION

LENS	GUIDING QUESTION	PROPOSED EVIDENCE
Verification	Did we build the system according to the requirements and architectural rules?	Unit/integration assertions, schema checks, security regression, coverage and CI artifacts.
Validation	Did we build the right experience for real users?	Role-based UAT scripts, observed workflows, participant sign-off and defect log.
Release decision	Is remaining risk acceptable for demonstration or deployment?	Traceability, critical-case status, accepted NFR evidence and open-defect decision.

IN AND OUT OF SCOPE

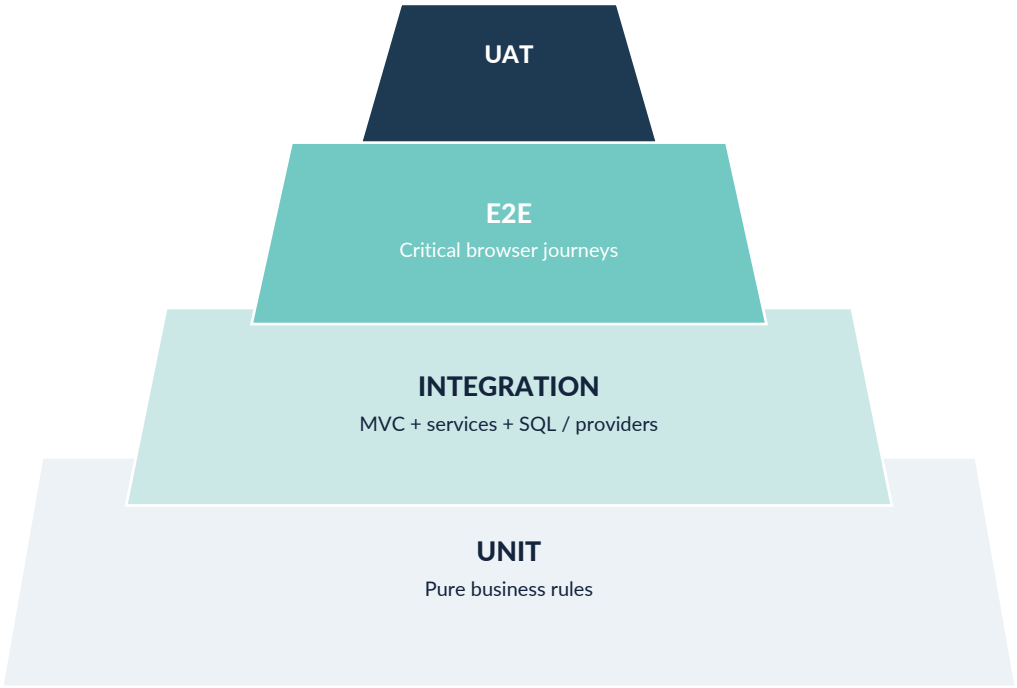
- In scope: proposed test cases, environments, fixtures, entry/exit criteria, traceability, defect handling, reporting and UAT procedures.
- Out of scope: claims of executed tests, screenshots of a real running EduHub build, actual load metrics, live Stripe charges or approval of proposed requirements.

> Dependency This plan is ready for implementation, but test execution requires an EduHub build, a provisioned sandbox/staging environment, test users, and an approved requirements baseline.

Quality strategy at a glance

Risk-led testing anchored in the N-Tier modular-monolith architecture

TEST LAYERS



Interpretation The shapes communicate relative automation emphasis, not a measured test-count or coverage distribution. The working mix is many fast unit checks, fewer realistic integration checks, a selected set of E2E journeys and stakeholder UAT.

PRIORITIZATION

- P0 / high risk: authentication, cross-role isolation, Stripe verification, enrollment fulfillment, certificate eligibility, course moderation and protected media.
- P1 / core value: search, authoring, progress, quizzes, cart pricing, instructor interaction and analytics.
- P2 / presentation polish: optional sharing paths, chart aesthetics and convenience features, subject to the approved delivery scope.

Quality gates & test ownership

Candidate release controls and accountable evidence owners

GATES FROM COMMIT TO ACCEPTANCE

GATE	ENTRY / EVIDENCE	CANDIDATE PASS DECISION
G0: Design review	Source requirements + technical design reviewed	All high-risk decisions have test oracle and named owner.
G1: Build & unit	Compile, lint/static analysis, 14 high-value unit cases + expanded suites	No build errors; required pure-rule cases pass.
G2: Integration	Real SQL Server test instance, isolated Stripe/Azure sandbox/fakes	All payment, role, persistence and retry regressions pass.
G3: E2E + NFR	Browser smoke, performance and security evidence, accessibility checks	Mandatory journeys pass; candidate NFR thresholds assessed.
G4: UAT	Seeded staging, actor scripts, captured results	Stakeholder accepts core student, instructor and admin journeys.
G5: Release review	Defect summary + traceability + deployment smoke	No open blocker/critical defects; exceptions documented and approved.

SUGGESTED OWNERSHIP

ROLE	ACCOUNTABILITY
Test lead	Own plan, traceability, daily test status, defect triage and evidence quality.
Backend engineer	Write service/unit/integration cases and SQL/provider fixtures.
Frontend engineer	Create browser E2E and accessibility checks for MVC/Razor UI.
DevOps owner	Provision isolated staging, secrets, CI jobs, data reset and load telemetry.
UAT stakeholders	Confirm role-specific acceptance and business expectations.
Supervisor	Approves changed requirements, risk acceptance and final demonstration gate.

Toolchain & automation harness

Recommended tools match the previously selected .NET / Azure technology stack

TOOLING DECISION REGISTER

AREA	RECOMMENDED TOOL	PURPOSE / IMPLEMENTATION NOTE
Unit	xUnit, dotnet test	Deterministic service and domain-rule test projects. [R1]
Assertions / fakes	xUnit Assert; test doubles via interfaces	Mock Stripe, Blob, clock, mail and repositories only at intended seams.
Coverage	Coverlet + ReportGenerator	Produce Cobertura report and human-readable coverage. [R2]
MVC integration	WebApplicationFactory + TestServer	Host actual ASP.NET Core composition with test configuration. [R3]
Relational integration	Disposable SQL Server / Azure SQL test DB	Verify migrations, FK/unique constraints, transactions and provider behavior. [R4]
Browser E2E	Playwright .NET	Cross-browser critical journeys and screenshots/traces on failure. [R5]
Payment contract	Stripe test mode + CLI / test event fixture	Check signed webhook handling without live payment exposure. [R6]
Performance	k6 or Azure Load Testing (proposed)	Run reproducible workloads, measure p50/p95 and error rates.
Security	OWASP ASVS checklist + targeted dynamic checks	Privilege, injection, CSRF, secret exposure and safe error flows. [R7]
Accessibility	axe-core + manual keyboard / screen-reader sampling	Verify WCAG 2.2 AA core paths. [R8]

Test project layout Suggested projects: EduHub.Tests.Unit, EduHub.Tests.Integration, EduHub.Tests.E2E. Keep these separate from the three production assemblies and ensure no test-only credentials ship with the web app.

Test environments & data

Reproducible, isolated fixtures for high-fidelity scenarios

ENVIRONMENT MATRIX

ENV	DATA & PROVIDERS	PURPOSE / GUARDRAIL
Local Dev	Disposable DB; fake Stripe/Blob; local seed	Fast unit/integration developer feedback; no real customer data.
CI Integration	SQL Server test service; faked external adapters	Deterministic assertions on real EF Core migrations and constraints.
QA/Staging	Azure App Service + isolated Azure SQL & Blob; Stripe test mode	E2E, provider contract, manual UAT, accessibility and load tests.
Production	Live providers; synthetic read-only probes only	Deployment verification and monitoring; never run destructive test scripts.

SEEDED DATASET BASELINE - PROPOSED

SET	DATA REQUIRED	WHY
Identity	Guest, 2 Students, 2 Instructors, 1 Admin	Validate ownership and role isolation.
Courses	Draft, Submitted, Approved, Rejected, Free, Paid	Catalog and moderation state coverage.
Learning	Sections, media, quizzes, 2 enrolled/non-enrolled users	Progress and protected video checks.
Commerce	Two carts, paid order, coupon valid/expired, Stripe event ids	Pricing, retries and order consistency.
Certificates	Eligible and ineligible enrollments; known verification code	Award guards and public verification.
Telemetry	Stable sample requests and deterministic timestamps	Reproducible quality and analytics data.

Cleanup and determinism Fixture IDs must be unique per run; freeze time for coupon-expiry tests; isolate workers; reset database by rollback or re-seed. Store Stripe secrets and Azure credentials in protected CI/staging settings, never source control.

Unit testing / pure rules

14 reference unit test cases; this is a minimum candidate catalog

UNIT TEST MATRIX

ID	MODULE / CONDITION	DETERMINISTIC EXPECTATION
UT-01	Pricing - Percentage discount	100.00 with 15% -> 85.00; explicit currency rounding
UT-02	Pricing - Fixed-amount discount	Floor at zero; no negative payable total
UT-03	Pricing - Coupon eligibility	Expired, usage-cap, course mismatch -> denied
UT-04	Pricing - Multi-item cart subtotal	Sum eligible items once; repeat additions do not inflate total
UT-05	Learning - Progress percentage	2 of 5 lessons -> 40%; 0 lessons handled safely
UT-06	Learning - Duplicate progress event	Replaying lesson complete does not increment count twice
UT-07	Assessments - Multiple-choice grading	Correct option -> weight; wrong -> 0

EXECUTION BOUNDARY

Use service/domain classes with fake clock and interface-based stubs. Avoid networking, UI rendering or database setup in these checks. Table rows are test specifications, not reports of actual execution.

> Coverage target The previous SRS proposes at least 80% statement coverage for pricing, grading, progress, and certificate-eligibility service modules (NFR-25). Coverage is evidence of exercised paths, not a guarantee of behavioral correctness.

Unit testing / guards & events

Boundary values, retry conditions and implementation-independent oracles

UNIT TEST MATRIX - CONTINUED

ID	MODULE / CONDITION	DETERMINISTIC EXPECTATION
UT-08	Assessments - True/false grading	Known true/false values scored deterministically
UT-09	Assessments - Fill-blank grading	Documented comparison/normalization policy applied
UT-10	Assessments - Passing threshold	Score >= configured threshold passes; boundary tested
UT-11	Certificates - Issuance eligibility	All required lessons/quizzes satisfied -> eligible
UT-12	Certificates - Ineligible certificate	Any outstanding required component -> not eligible
UT-13	Courses - Publish guard	Draft lacking required fields stays unpublishable
UT-14	Notifications - Observer behavior	Repeated signal yields one business intent event

DESIGN EXAMPLES

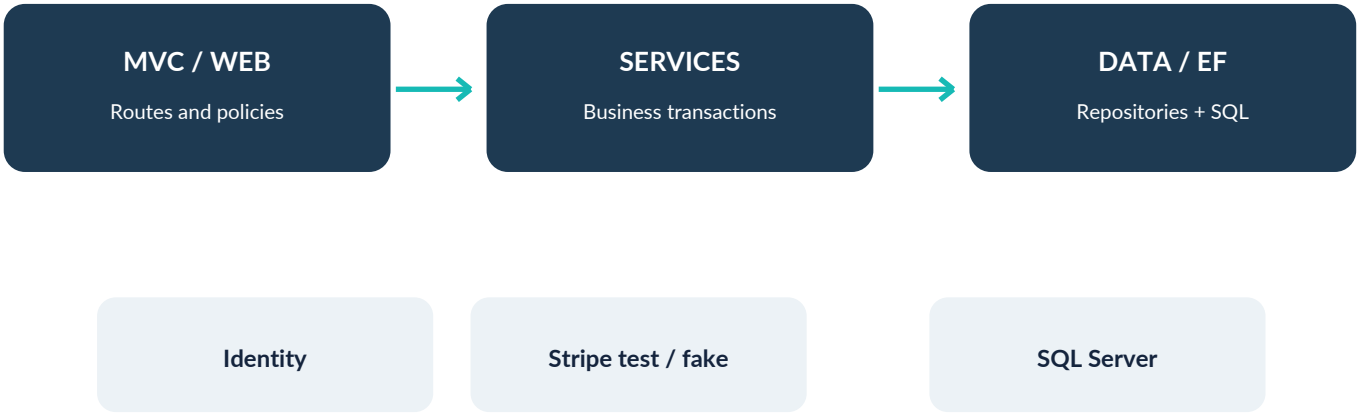
- Boundary values: zero, one, and multiple lessons; 0%, exactly passing mark, 100% quiz scores; positive and zero payable amounts.
- Equivalence classes: active/expired coupon; enrolled/not enrolled; correct/incorrect answer; draft/approved/rejected course.
- Property invariants: 0 <= progress <= 100; payable total >= 0; duplicate event is effect-free; owner requirement always enforced.

Recommendation Run these cases on every pull request. Unit tests should not assert EF Core SQL translation; those behaviors belong in SQL Server integration tests. [R4]

Integration test design

Verify composition: MVC controller -> service -> data / external adapter

INTEGRATION TOPOLOGY



TEST ISOLATION AND REALISM

BOUNDARY	TEST DOUBLE OR REAL SERVICE	REASON
Controller -> service	Real MVC host, replace only configured external adapter	Verify routes, auth, antiforgery and DI registrations.
Service -> EF Core	Real SQL Server relational instance	Catch transaction, FK, collation and LINQ provider differences.
Stripe checkout/webhook	Contract fixture and Stripe test environment	Prove signature, idempotency and event semantics.
Blob + media pipeline	Fake adapter in CI; isolated Blob in staging	Exercise unhappy paths without paid storage dependence.
Async notification	Fake sink with recorded intent	Verify post-commit dispatch and duplicate suppression.

Provider fidelity EF Core explicitly warns that its in-memory provider is not a substitute for SQL Server transaction/query tests. Unit test repositories with doubles, but run persistence and migrations against SQL Server. [R4]

Functional case catalog / A

TC-001 through TC-008 | Identity, discovery and enrollment

CASE	ACTIVITY / INPUT	EXPECTED RESULT	UC
TC-001	Register permitted learner/instructor account	Exactly one account/profile; invalid duplicate rejected	UC-01
TC-002	Sign in, sign out and access controlled page	Valid session starts; sign-out blocks protected route	UC-02
TC-003	Search and filter published course catalog	Only matching published courses returned with correct paging	UC-03
TC-004	Open course details as guest	Correct public metadata; draft hidden	UC-04
TC-005	Verify certificate by public unique code	Issued certificate shown; unknown code returns safe result	UC-05
TC-006	Edit profile and saved preferences	Owner change persists; cross-user edit denied	UC-06
TC-007	Enroll in a free published course	Single enrollment created without Stripe payment	UC-07
TC-008	Add/remove/update several cart items	Cart and computed subtotal consistent	UC-08

Specification convention Every test case is a proposed test design and remains NOT RUN. Expected results refer to observable response and persisted state where applicable.

Functional case catalog / B

TC-009 through TC-016 | Checkout, protected learning and grading

CASE	ACTIVITY / INPUT	EXPECTED RESULT	UC
TC-009	Apply valid/expired/invalid coupon	Price uses allowed strategy; invalid discount rejected	UC-09
TC-010	Create paid checkout session	Order pending; server-computed total matches Stripe session	UC-10
TC-011	Read completed order and receipt	Only owner views order; receipt matches settled values	UC-11
TC-012	Open an enrolled course	Entitled student admitted; outsider denied	UC-12
TC-013	Play protected video lesson	Authorized signed asset URL and functional player	UC-13
TC-014	Mark lesson as complete	Student-scoped idempotent progress update	UC-14
TC-015	Display course progress percentage	Aggregate equals seeded completed/eligible lesson ratio	UC-15
TC-016	Take quiz and view grade	Answer types scored; explanations and pass result correct	UC-16

Functional case catalog / C

TC-017 through TC-024 | Learner engagement and award use

CASE	ACTIVITY / INPUT	EXPECTED RESULT	UC
TC-017	Save timestamped lesson note	Note saved only on owned enrolled lesson	UC-17
TC-018	Bookmark/unbookmark lesson	Single scoped bookmark toggles correctly	UC-18
TC-019	Submit course question	Question belongs to target course and visible appropriately	UC-19
TC-020	Read authorized course announcements	Latest announcements ordered and visible to enrollee	UC-20
TC-021	Submit rating and written review	Valid 1-5 rating and review saved with allowed ownership	UC-21
TC-022	Vote existing review helpful	Vote idempotency and count consistency	UC-22
TC-023	Download issued certificate PDF	PDF fields/code match eligible student + course	UC-23
TC-024	Share certificate on LinkedIn	Share link contains expected public verification target	UC-24

Functional case catalog / D

TC-025 through TC-032 | Course builder and content lifecycle

CASE	ACTIVITY / INPUT	EXPECTED RESULT	UC
TC-025	Create and edit owned draft course	Draft changes persist; foreign owner denied	UC-25
TC-026	Arrange sections and lessons	Ordered hierarchy preserved and invalid reorder blocked	UC-26
TC-027	Upload course media to protected storage	Object stored with expected metadata; public read denied	UC-27
TC-028	Bulk upload selected assets	Per-file successes and validation errors surfaced	UC-28
TC-029	Set list price and course discount	Pricing and promotion stored for owner only	UC-29
TC-030	Preview instructor draft course	Owner preview available without public exposure	UC-30
TC-031	Submit course for approval	Eligible draft enters moderation queue	UC-31
TC-032	Modify/retire content safely	Changes honor approvals and existing enrollments	UC-32

Functional case catalog / E

TC-033 through TC-040 | Instructor / Admin governance

CASE	ACTIVITY / INPUT	EXPECTED RESULT	UC
TC-033	Author questions and passing rules	Quiz composition and validation persist	UC-33
TC-034	Read course enrollment and earnings stats	Only owner-specific aggregates returned	UC-34
TC-035	Answer enrolled student question	Reply linked to question; nonowner cannot answer	UC-35
TC-036	Post instructor course announcement	Visible to expected enrolled audience	UC-36
TC-037	Admin inspect submitted course	Admin can view queued content; non-admin denied	UC-37
TC-038	Admin approve or reject course	Status and public catalog visibility change correctly	UC-38
TC-039	Admin manage users and roles	Authorized changes only; public Admin assignment blocked	UC-39
TC-040	Admin manage categories and tags	Uniqueness and referenced course relationships preserved	UC-40

Functional case catalog / F

TC-041 through TC-048 | Reporting, providers, events and certificates

CASE	ACTIVITY / INPUT	EXPECTED RESULT	UC
TC-041	Open platform analytics dashboard	Seeded totals align to orders/enrollments/progress	UC-41
TC-042	Update platform configuration	Auditable authorized configuration update	UC-42
TC-043	Receive Stripe payment event	Signature and event-type verified; duplicates recorded safely	UC-43
TC-044	Fulfill confirmed paid order	One payment effect and one enrollment per course	UC-44
TC-045	Generate eligible certificate	One valid verifiable certificate, replays do not duplicate	UC-45
TC-046	Process media and create thumbnail	Media state transitions on success/failure, retry-safe	UC-46
TC-047	Authorize enrolled video access	Valid short-lived access token; non-entitled user denied	UC-47
TC-048	Send event-triggered notification	One relevant notification; core action survives delivery failure	UC-48

Cross-system integrity TC-043 and TC-044 must inspect actual order/payment/enrollment rows. A successful browser redirect alone does not establish paid entitlement; fulfillment requires trusted server-side payment verification. [R6]

Negative-path catalog / 1

TC-049 through TC-056 | Security, reliability and failure paths

ID / LAYER	TRIGGER	EXPECTED OBSERVABLE RESULT	TRACE
TC-049 / SEC	Forged role field in registration	No privileged role created; access rejected	UC-01 / NFR-03
TC-050 / SEC	Instructor B edits Instructor A draft	403 or safe not-found; unchanged DB	UC-25 / NFR-03
TC-051 / SEC	Direct media endpoint / copied URL without entitlement	No durable unauthorized access; URL expires	UC-47 / NFR-03
TC-052 / SEC	Tamper JSON or stripe-signature	No settlement, enrollment or certificate	UC-43 / NFR-06
TC-053 / INT	Replay same verified Stripe event	One durable event effect; no duplicate enrollment	UC-43 / NFR-16
TC-054 / INT	Two workers process same paid order	Uniqueness/transaction assures one enrollment	UC-44 / NFR-18
TC-055 / INT	Visit success URL without paid webhook	Order remains pending; access denied	UC-10 / BR-04
TC-056 / UNIT+INT	Expired or restricted coupon applied twice	No excess discount; totals recomputed server-side	UC-09 / FR-048

Exit expectation These cases have high risk exposure: a failure may reveal private data, create duplicate financial effects, bypass enrollment or issue an invalid credential. Track with P0/P1 priority even if the feature works on the happy path.

Negative-path catalog / 2

TC-057 through TC-064 | Security, reliability and failure paths

ID / LAYER	TRIGGER	EXPECTED OBSERVABLE RESULT	TRACE
TC-057 / SEC	Public catalog URL for unapproved draft	Draft not indexed or accessible to guest	UC-03 / FR-013
TC-058 / SEC	Oversized, wrong MIME, malformed upload	Rejected safely; no public exposed asset	UC-27 / NFR-04
TC-059 / INT	Transcoder fails mid-workflow	Asset not published; recoverable status	UC-46 / NFR-17
TC-060 / SEC	Student modifies another enrollment progress	Denied; other student record unchanged	UC-14 / BR-05
TC-061 / INT	Repeated completion actions or delayed retry	Progress bounded, no duplicate certificate	UC-14 / NFR-16
TC-062 / SEC	Override submitted score or correct-answer field	Server regrades; attempt score valid	UC-16 / NFR-04
TC-063 / INT	Incomplete lesson or failed mandatory quiz	No certificate generated	UC-45 / FR-063
TC-064 / INT	Repeat award event concurrently	Unique verified code, single certificate	UC-45 / FR-082

Exit expectation These cases have high risk exposure: a failure may reveal private data, create duplicate financial effects, bypass enrollment or issue an invalid credential. Track with P0/P1 priority even if the feature works on the happy path.

Negative-path catalog / 3

TC-065 through TC-072 | Security, reliability and failure paths

ID / LAYER	TRIGGER	EXPECTED OBSERVABLE RESULT	TRACE
TC-065 / SEC	Post state-changing form without antiforgery token	Rejected; no state mutation	UC-25 / NFR-04
TC-066 / ACC	Navigate forms/dialog/course builder via keyboard	Usable focus path with labeled errors	UC-16 / NFR-19
TC-067 / SEC	Trigger Stripe/Identity exception with sensitive strings	Logs omit tokens, payment and credentials	UC-43 / NFR-07
TC-068 / INT	Apply migrations on fresh/upgrade database	Constraints/indexes present; data preserved	UC-25 / NFR-30
TC-069 / INT	Simulate Stripe/Azure transient timeout	Controlled error and safe retry; no partial commit	UC-43 / NFR-31
TC-070 / INT	Replay vote same user/review	One vote persisted; count consistent	UC-22 / FR-062
TC-071 / SEC	Manipulate order ID with another session	No disclosure of payment history	UC-11 / NFR-03
TC-072 / SEC	Client supplies negative total or discount	Server price authoritative; bad total rejected	UC-10 / FR-050

Exit expectation These cases have high risk exposure: a failure may reveal private data, create duplicate financial effects, bypass enrollment or issue an invalid credential. Track with P0/P1 priority even if the feature works on the happy path.

Risk-based execution order

Use failure impact and likelihood to focus scarce graduation-project testing time

CANDIDATE DOMAIN RISK SCORE



Interpretation Scores are qualitative proposed priorities, not findings from executed tests. P0 areas require both happy-path and adversarial evidence before release recommendation. Re-rank when implementation details or actual defects change exposure.

Case studies / payment integrity

High-risk, stepwise verification procedures and precise pass oracles

01 / VERIFIED WEBHOOK AND FULFILLMENT

Scenario reference TC-043 + TC-044 | UC-43/44 | FR-054/055, NFR-06/16/18

Fixture and preconditions: Isolated SQL Server; pending order with two paid courses; verified signed Stripe test event; empty enrollment set.

STEP	ACTION / ASSERTION
1	Create pending order and persist calculated line totals.
2	Submit signed completed payment event to webhook handler.
3	Inspect event record, order state, payment and enrollment rows.
4	Submit the same event twice; issue two concurrent deliveries.

Pass oracle Exactly one settled business effect; exactly one enrollment per purchased course; no access granted by browser redirect alone.

Failure mode covered: Duplicate purchase, double enrollment and forged payment.

Case studies / media controls

Authorization and resiliency across Azure Blob and media processing

02 / SECURE VIDEO AUTHORIZATION

Scenario reference TC-047 + TC-051 | UC-47 | FR-028, NFR-03/32

Fixture and preconditions: Approved course with ready video; enrolled Student A; outsider Student B; media configured private.

STEP	ACTION / ASSERTION
1	Request video for A through MVC authorization route.
2	Inspect signed URL expiry and bounded scope.
3	Request same asset as B and with unauthenticated session.
4	Try previously issued URL after expiry.

Pass oracle Only A obtains time-limited access; outsider is denied; asset container is not anonymous; expiry enforced.

Failure mode covered: Unauthorized streaming by guessing lesson IDs or reusing links.

Case studies / completion rules

Avoid awarding a certificate based solely on progress-bar presentation

03 / ELIGIBILITY AND IDEMPOTENT AWARD

Scenario reference TC-045 + TC-063/064 | UC-45 | FR-063..067, FR-082

Fixture and preconditions: Student completed all lessons except one; mandatory quiz initially below threshold; no certificate record.

STEP	ACTION / ASSERTION
1	Attempt issuance while completion incomplete.
2	Complete final lesson but keep quiz failure.
3	Pass required quiz; trigger completion event.
4	Replay the same event and run concurrent issuance workers.
5	Verify public code and inspect generated PDF.

Pass oracle No award until all rules pass; then exactly one certificate record, one code and correct PDF identity fields.

Failure mode covered: Premature or duplicate certificate issuance.

Case studies / governance

Course-state permissions and visibility changes

04 / CONTENT MODERATION AND PUBLISH

Scenario reference TC-031 + TC-038 + TC-057 | UC-31/38 | FR-013/023/070

Fixture and preconditions: Instructor owns a complete draft; Admin reviewer and unrelated Student account available.

STEP	ACTION / ASSERTION
1	Instructor submits draft into review queue.
2	Student queries public catalog for submitted course.
3	Admin inspects and approves course.
4	Student repeats search and opens course details.
5	Submit an incomplete second course and reject it.

Pass oracle Submitted/rejected drafts are not public; approved course becomes public; non-admin cannot approve.

Failure mode covered: Publication before quality approval and role escalation.

Case studies / authentication

Direct object access, role injection and session lifecycle

05 / IDENTITY AND OWNERSHIP REGRESSION

Scenario reference TC-001/002 + TC-049/050 | UC-01/02/25 | NFR-02/03

Fixture and preconditions: User registration endpoint, two instructor accounts, Admin-only route, known foreign draft course.

STEP	ACTION / ASSERTION
1	Register user while tampering form role with Admin value.
2	Inspect persisted role assignments using trusted query.
3	Attempt Admin-only route as registered user.
4	Attempt edit of second instructor course.
5	Sign out and retry previously protected route.

Pass oracle Public Admin role not granted; unauthorized requests rejected; no foreign data changed; sign-out invalidates access.

Failure mode covered: Privilege escalation, IDOR and stale authentication.

Case studies / grading

Quiz scoring, boundary passing and tampered score payloads

06 / GRADE AND PROGRESS CORRECTNESS

Scenario reference UT-07..10 + TC-016/062 | UC-16 | FR-038..044

Fixture and preconditions: Seed quiz with MCQ, true/false and fill-blank questions, published solutions and known score weights.

STEP	ACTION / ASSERTION
1	Submit answer set just below pass threshold.
2	Submit answer set exactly at threshold.
3	Submit answer set above threshold.
4	Tamper client payload to include an arbitrary higher grade.
5	Inspect stored attempt, feedback and retake policy.

Pass oracle Server-computed scores match expected weights and pass boundary; forged grade ignored; attempt policy enforced.

Failure mode covered: Client-trusted results and inconsistent grade rounding.

Case studies / schema & reports

SQL Server fidelity and reproducible data aggregation

07 / MIGRATION AND ANALYTICS CORRECTNESS

Scenario reference TC-068 + TC-041 | UC-41 | NFR-30, FR-073..077

Fixture and preconditions: Clean SQL Server test database, previous schema snapshot, seeded settled order/learning records.

STEP	ACTION / ASSERTION
1	Apply baseline EF Core migrations on empty DB.
2	Verify FK/unique/check constraints and key indexes.
3	Apply forward migration against seeded prior schema.
4	Compare Admin report totals against independent SQL aggregates.

Pass oracle No failed migrations or lost seeded data; analytics aggregates match controlled financial and course events.

Failure mode covered: Provider-specific schema differences, double-counted revenues.

Case studies / resilient events

Do not roll back committed business outcomes due to notification failures

08 / NOTIFICATION AND MEDIA FAILURE CONTAINMENT

Scenario reference TC-046/048 + TC-059/069 | UC-46/48 | NFR-17/31

Fixture and preconditions: Simulate Blob processing timeout, failed notification provider and recorded retry queue.

STEP	ACTION / ASSERTION
1	Trigger a legitimate lesson completion event.
2	Cause outbound notification to fail transiently.
3	Check progress commit and safe retry intent.
4	Cause video transcoding to fail; inspect media status.
5	Retry processing without duplicating storage references.

Pass oracle Core transaction remains committed; failure is observable and retry-safe; bad media is not published.

Failure mode covered: Partial transactions and unbounded retries.

Non-functional validation / 1

Security / performance | proposed quality requirements NFR-01 to NFR-08

REQ	PROPOSED VERIFICATION / ORACLE	EVIDENCE
NVT-01	Encrypted transport - All application pages, APIs and protected asset delivery shall use HTTPS with valid certificates.	Check public endpoints and media redirects; no protected operation is served over plain HTTP.
NVT-02	Identity safeguards - Authentication shall use ASP.NET Core Identity password hashing, lockout and session protections.	Identity conformance review plus negative authentication tests.
NVT-03	Resource-level authorization - Protected records and media shall be authorized by role AND resource ownership/enrollment.	Cross-user/cross-instructor access tests return denied without data leakage.
NVT-04	CSRF and input safety - State-changing MVC forms shall validate anti-forgery tokens; user input shall be validated and output encoded.	Anti-forgery and injection/XSS regression test suite passes.
NVT-05	Secret custody - Stripe keys, storage credentials and connection secrets shall not be committed to the source repository.	Automated secret scan and production configuration review show zero embedded secrets.
NVT-06	Payment confidentiality - EduHub shall not store raw card numbers or CVV and shall validate Stripe webhook signatures.	Database audit finds no card credentials; invalid signatures are rejected.
NVT-07	Minimal personal exposure - Public verification and log outputs shall reveal only intended public certificate fields and redacted sensitive data.	Inspect public certificate response and application logs for prohibited identifiers.
NVT-08	Abuse resilience - Login, coupon and public verification endpoints shall have bounded retry/rate control mechanisms.	Automated burst tests reach configured limits without privileged bypass.

Non-functional validation / 2

Performance / reliability | NFR-09 to NFR-16

REQ	PROPOSED VERIFICATION / ORACLE	EVIDENCE
NVT-09	Public page response - At 50 concurrent test users, p95 server response time for catalog and course-detail requests shall be <= 2.5 s.	k6/JMeter load test for 10 minutes on documented seeded Azure environment.
NVT-10	Dashboard response - At 30 concurrent test users, p95 response time for dashboard summaries shall be <= 3.0 s.	Repeatable load test with fixed database size and telemetry.
NVT-11	Video playback start - With a 10 Mbps test connection and ready media, p95 time to first playable frame shall be <= 5 s.	Browser playback test under controlled network conditions; exclude user pause.
NVT-12	Bounded result sets - Catalog, report and history list endpoints shall use bounded pagination or equivalent limits.	High-volume seed test shows no unbounded query result payloads.
NVT-13	Concurrent operation - The solution shall support 50 concurrent representative sessions with <=1% application-level failed operations.	Mixed-workload 10-minute load test; integration sandbox responses reported separately.
NVT-14	Proposed service availability - The hosted web application shall target >=99.5% availability over a defined 30-day observation window.	Use synthetic HTTPS availability checks; exclude agreed maintenance windows.
NVT-15	Backup and restore - Database recovery procedures shall target RPO <=24 h and RTO <=8 h for the capstone environment.	Conduct and time a documented restore drill using a representative backup.
NVT-16	Exactly-once business effect - At-least-once external event delivery shall not duplicate purchases, enrollments or certificates.	Replay a processed payment/completion event and confirm unchanged row counts.

Non-functional validation / 3

Resilience / accessibility / compatibility | NFR-17 to NFR-24

REQ	PROPOSED VERIFICATION / ORACLE	EVIDENCE
NVT-17	Safe retry after outage - Transient storage, email and payment processing failures shall be retried safely or marked for intervention.	Inject provider timeouts; verify durable failure state and no partial access grants.
NVT-18	Transaction integrity - Order, enrollment, progress and certificate updates shall preserve referential and business-rule consistency.	Induce write failures mid-flow and verify no invalid committed partial state.
NVT-19	Accessibility target - The web interface shall target WCAG 2.2 Level AA for primary learner, instructor and Admin journeys.	Automated audit plus manual keyboard/screen reader checks against defined AA criteria.
NVT-20	Keyboard operability - Navigation, forms, dialogs, quiz actions and course builder controls shall be keyboard operable.	Walk core scenarios without pointer and confirm visible focus.
NVT-21	Responsive layout - Core pages shall render without horizontal content clipping at widths from 320 px to desktop.	Responsive viewport audit at 320, 768 and 1440 px for representative pages.
NVT-22	Accessible feedback - Validation errors and completion outcomes shall be expressed in clear text associated with the relevant control.	Invalid-submission tests verify text labels and screen-reader announcements.
NVT-23	Browser coverage - Critical flows shall operate on current supported Chrome, Edge and Firefox desktop versions.	Cross-browser smoke suite passes on versions recorded at UAT time.
NVT-24	N-Tier boundaries - Controllers shall depend on application-service contracts, not directly on EF Core repositories/DbContext.	Architecture dependency test and solution reference review show allowed edges only.

Non-functional validation / 4

Maintainability / deployments / media delivery | NFR-25 to NFR-32

REQ	PROPOSED VERIFICATION / ORACLE	EVIDENCE
NVT-25	Testable business logic - Pricing, grade evaluation, progress and certificate eligibility logic shall be covered by automated unit tests.	Minimum 80% statement coverage on those named service modules (proposal).
NVT-26	Integration regression - Checkout webhook, role authorization and persistence workflows shall have repeatable integration tests.	CI test suite passes with sandbox/fake external dependencies.
NVT-27	Structured diagnostics - The application shall emit structured logs with correlation IDs and sanitized exception information.	An end-to-end request can be traced across controller, service and data activities.
NVT-28	Environment configuration - Environment-dependent settings shall be injected/configured without modifying compiled application code.	Deploy to test/prod settings with the same build artifact and no source edits.
NVT-29	Platform deployment - The application shall be deployable on Azure App Service with its configured Azure dependencies.	Clean staging deployment passes startup, login, catalog and storage smoke tests.
NVT-30	Schema migration - EF Core migrations shall apply cleanly to an empty schema and preserve versioned schema changes.	Fresh database migration and forward deployment rehearsal succeed.
NVT-31	Provider isolation - Stripe and Azure storage failures shall be surfaced as controlled application outcomes, not unhandled crashes.	Forced provider outage returns an actionable error and maintains transactional invariants.
NVT-32	Media delivery scaling - Video byte delivery shall use Blob/CDN paths rather than proxying large streams through the MVC application.	Browser network inspection verifies media served from the media delivery endpoint.

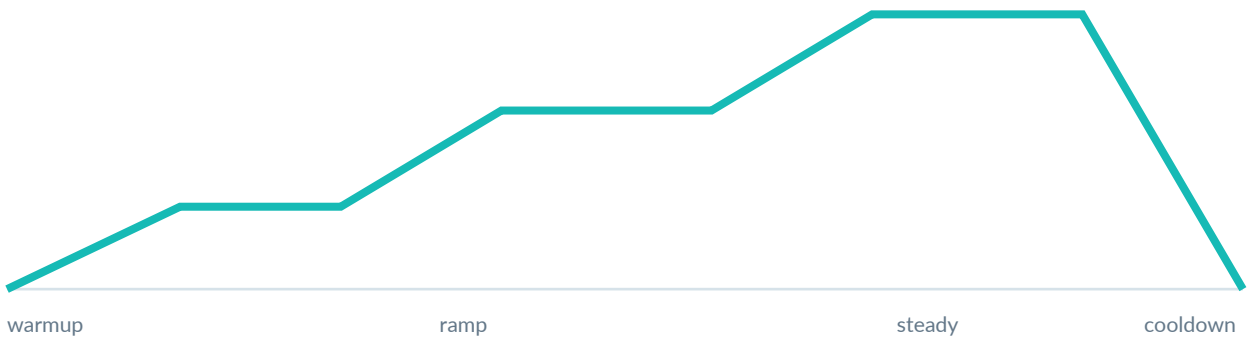
Performance & load protocol

Repeatable workload design for the SRS candidate thresholds

SPECIFIED CANDIDATE TARGETS FROM THE PRIOR SRS

REQ	MEASURE / CONDITION	TARGET
NFR-09	Catalog/course detail, 50 concurrent test users, 10 min	p95 <= 2.5 s
NFR-10	Dashboard, 30 concurrent users, fixed seed	p95 <= 3.0 s
NFR-11	Video first playable frame, ready media, 10 Mbps connection	p95 <= 5.0 s
NFR-13	50 concurrent representative sessions	<= 1% application errors
NFR-14	30-day hosted observation window (availability is not a short load test)	>= 99.5%
NFR-15	Capstone backup/restore tabletop + timed exercise	RPO <= 24 h; RTO <= 8 h

PROPOSED LOAD SHAPE



Test controls Fix dataset size, scenario mix, region, SKU, browser/network profiles and recording interval. Save raw latency percentiles, failure counts, resource utilization and scenario traces. Do not report a pass before recording measured evidence.

Security & accessibility protocol

Threat-driven and inclusive validation beyond nominal happy paths

SECURITY FOCUS AND PRIMARY CHECKS

AREA	THREAT / CONTROL	CANDIDATE EVIDENCE
Identity	Registration role tampering, credential abuse, stale sessions	Role DB query, denied response, lockout scenario.
Authorization	IDOR on course, enrollment, media and receipts	Foreign-ID negative requests across users.
Input safety	CSRF, forged totals, output encoding, upload validation	Rejected mutation; escaped output; accepted media limits.
Payments	Invalid Stripe signatures and webhook replay	No settlement from forged event; one effect from replay.
Confidentiality	Secrets in logs, public cert metadata, media URL expiry	Redacted diagnostics and minimal public fields.
Availability	Retry storms, rate-limit bypass, provider timeout	Bounded retry and controlled failure outcome.

ACCESSIBILITY: WCAG 2.2 AA TARGET

- Keyboard: navigate catalog, checkout, player controls, quiz and course builder without trapping focus.
- Forms: programmatic labels, validation associations, instructions and visible focus indicator.
- Content: color contrast, semantic headings, player subtitles/captions as agreed, responsive zoom and error announcement.

Standard usage OWASP ASVS 5.0 offers a structured application-security checklist [R7]. WCAG 2.2 defines testable accessibility success criteria [R8]. An automated scanner alone is not a WCAG AA conformance claim.

User acceptance test charter

Product validation with learners, instructors and administrators

UAT APPROACH

DIMENSION	APPROACH
Participants	At least one trained representative each for Student, Instructor and Admin; Guest tasks handled by observer or Student.
Environment	Seeded staging using Stripe test mode and isolated media and SQL stores; browser versions documented.
Method	Participants complete scenario with minimal prompting, then judge business outcome and usability.
Recorder	Capture actual result, deviations, screenshot/recording, severity and participant comments.
Acceptance rule	All P0 UAT scripts must pass; any rejected workflow requires fix/retest or explicit supervisor risk acceptance.
UAT output	Signed/dated checklist, open issues, exceptions and demonstration readiness recommendation.

WHY THIS IS NOT A SIMULATED SIGN-OFF

The table below is a proposed UAT charter only. All UAT scripts currently carry status NOT RUN because no real user session, staging build, screenshots or acceptance decision has been supplied.

> Team preparation Confirm accessibility needs, role credentials, environment URL, timezone/currency, planned certificate wording and allowed test payment methods before UAT starts.

UAT scripts / learner journeys

UAT-01 to UAT-06 | Guest / Student

ID / ACTOR	TASK	ACCEPTED OBSERVABLE OUTCOME	UC
UAT-01 / Guest	Find a course - Search by category/level; open course preview	Only published matching course shown	UC-03/04
UAT-02 / Student	Start free course - Enroll in free course, open first lesson	Immediate entitlement with no checkout	UC-07/12
UAT-03 / Student	Buy paid courses - Add two paid courses; apply valid code; complete test checkout	Correct total, receipt, two entitlements	UC-08/09/10/11
UAT-04 / Student	Continue learning - Resume video; mark lesson done; write note/bookmark	Progress, note and bookmark persist	UC-13/14/15/17/18
UAT-05 / Student	Pass section quiz - Take seeded quiz at threshold, inspect feedback	Expected grade and required feedback	UC-16
UAT-06 / Student	Earn certificate - Complete learning and verify award publicly	PDF download and valid public code	UC-23/05

UAT scripts / management

UAT-07 to UAT-12 | Student / Instructor / Admin

ID / ACTOR	TASK	ACCEPTED OBSERVABLE OUTCOME	UC
UAT-07 / Student	Course feedback - Ask question, submit review, vote helpful	Correct visibility and rating	UC-19/21/22
UAT-08 / Instructor	Build a course - Create draft with section, lessons, quiz, media	Draft preview complete, public still hidden	UC-25/26/27/33
UAT-09 / Instructor	Publish workflow - Submit prepared course, wait for Admin review	Approved course becomes discoverable	UC-31/38
UAT-10 / Instructor	Manage learners - Answer question, announce update, inspect stats	Only own course data modified	UC-34/35/36
UAT-11 / Admin	Govern platform - Approve/reject content, manage category and user	Correct access and audit trail	UC-37/38/39/40
UAT-12 / Admin	Validate reports - Compare seeded sales/completion results to UI	Totals agree with controlled dataset	UC-41

Defects, evidence & triage

A repeatable path from observed failure to verified closure

SUGGESTED SEVERITY MODEL

SEVERITY	DEFINITION	EXAMPLE
S1 / blocker	Compromises access, payment integrity or core demonstration; no safe workaround	Forged event grants paid access or public Admin role.
S2 / major	Breaks an essential journey but not an exploit of critical integrity	Approved course cannot enroll learners.
S3 / moderate	Affects secondary journey with documented workaround	Review helpful count updated late.
S4 / minor	Cosmetic or low-impact usability deviation	Noncritical chart label misalignment.

DEFECT WORKFLOW



EVIDENCE MANIFEST PER EXECUTION

- Record build commit, environment, time, data/fixture version, test ID, actor, steps, expected/actual outputs, evidence link and responsible tester.
- For critical integrations capture normalized DB assertions, Stripe event id, correlation id, sanitized request/response and browser trace when applicable.

No invented execution metrics The test result dashboard should remain empty / NOT RUN until the team supplies real execution logs. Sample gate thresholds are planning assumptions, not observed system performance.

CI/CD verification pipeline

Stage assurance checks while keeping external-test costs proportionate

PROPOSED QUALITY PIPELINE

1. **Checkout + restore**
   Pinned .NET SDK; dependency scanning
2. **Build + static checks**
   Compile; architecture reference rules
3. **Unit tests**
   Fast domain checks and coverage collection
4. **Integration tests**
   SQL Server + WebApplicationFactory; provider fakes
5. **Staging deployment**
   Secret injection and database migrations
6. **E2E & smoke**
   Playwright + Stripe test provider + Blob
7. **NFR / UAT decision**
   Scheduled load/security; human acceptance

Recommended cadence Commit/PR: build, unit, critical integration. Nightly: complete integration + browser E2E. Pre-demo/release: scheduled NFR, security, migration rehearsal and UAT. Failed critical stages block promotion.

Traceability strategy

Requirement-to-test links are derived from the existing SRS, not guessed

MAPPING LOGIC

The previous SRS links each FR-001..FR-083 to one or more UC-01..UC-48. This test plan assigns TC-001..TC-048 to the corresponding use cases, and links each FR to the associated cases. The mapping is design coverage only; it does not imply a passing result.

COVERAGE REGISTRATION

83/83
FRS MAPPED

48/48
UCS MAPPED

32/32
NFR TESTS DESIGNED

TRACE EDGE	DEFINITION	HOW TO VALIDATE
FR -> UC	Previously documented SRS trace	Compare requirement ID and UC references.
UC -> TC	1:1 baseline named test case (TC-001..048)	Check expected result against use-case postconditions.
NFR -> NVT	NVT-01..32 correspond to SRS NFR-01..32	Measure against proposed numeric/qualitative fit criteria.
Risk -> negative	TC-049..072 add failure-oriented behavior	Run at least one adversarial path for each P0 risk.
UAT -> UC	UAT-01..12 validate cross-use-case business goals	Signed actor evidence and observed outcome.

Coverage accuracy Traceability coverage counts mean that IDs are assigned and specified. Test execution coverage remains 0 recorded runs in this document (all statuses NOT RUN).

FR-to-test matrix / 1

Prior SRS FR-001 through FR-021 - designed coverage, not execution proof

FR	REQUIREMENT TOPIC	SRS USE CASES	DESIGNED TEST CASES
FR-001	Account registration	UC-01	TC-001
FR-002	Sign-in and sign-out	UC-02	TC-002
FR-003	Role authorization	UC-02, UC-39	TC-002, TC-039
FR-004	Profile maintenance	UC-06	TC-006
FR-005	Administrator provisioning	UC-01, UC-39	TC-001, TC-039
FR-006	Protected actions	UC-07, UC-10, UC-25, UC-39	TC-007, TC-010, TC-025, TC-039
FR-007	Public course catalog	UC-03	TC-003
FR-008	Course search	UC-03	TC-003
FR-009	Course filters	UC-03	TC-003
FR-010	Category hierarchy	UC-03, UC-40	TC-003, TC-040
FR-011	Course detail view	UC-04	TC-004
FR-012	Visible course summaries	UC-03, UC-04	TC-003, TC-004
FR-013	Nonpublic drafts	UC-03, UC-31, UC-38	TC-003, TC-031, TC-038
FR-014	Create course drafts	UC-25	TC-025
FR-015	Edit owned courses	UC-25, UC-32	TC-025, TC-032
FR-016	Build sections	UC-26	TC-026
FR-017	Build lessons	UC-26	TC-026
FR-018	Draft and release actions	UC-25, UC-31	TC-025, TC-031
FR-019	Preview before release	UC-30	TC-030
FR-020	Update or retire content	UC-32	TC-032
FR-021	Bulk content upload	UC-28	TC-028

FR-to-test matrix / 2

Prior SRS FR-022 through FR-042 - designed coverage, not execution proof

FR	REQUIREMENT TOPIC	SRS USE CASES	DESIGNED TEST CASES
FR-022	Instructor pricing	UC-29	TC-029
FR-023	Administrative submission	UC-31, UC-37	TC-031, TC-037
FR-024	Store course videos	UC-27	TC-027
FR-025	Upload validation	UC-27	TC-027
FR-026	Video processing	UC-46	TC-046
FR-027	Thumbnail generation	UC-46	TC-046
FR-028	Secure media access	UC-47, UC-13	TC-047, TC-013
FR-029	Video lesson playback	UC-13	TC-013
FR-030	Free enrollment	UC-07	TC-007
FR-031	Paid enrollment access	UC-10, UC-12, UC-44	TC-010, TC-012, TC-044
FR-032	Enrollment authorization	UC-12, UC-13	TC-012, TC-013
FR-033	Lesson completion	UC-14	TC-014
FR-034	Course progress display	UC-15	TC-015
FR-035	Lesson notes	UC-17	TC-017
FR-036	Lesson bookmarks	UC-18	TC-018
FR-037	Create placement quizzes	UC-33	TC-033
FR-038	Supported question types	UC-33, UC-16	TC-033, TC-016
FR-039	Configure solutions	UC-33	TC-033
FR-040	Take a quiz	UC-16	TC-016
FR-041	Automatic scoring	UC-16	TC-016
FR-042	Instant feedback	UC-16	TC-016

FR-to-test matrix / 3

Prior SRS FR-043 through FR-063 - designed coverage, not execution proof

FR	REQUIREMENT TOPIC	SRS USE CASES	DESIGNED TEST CASES
FR-043	Pass threshold	UC-16, UC-45	TC-016, TC-045
FR-044	Retake policy	UC-16	TC-016
FR-045	Cart item management	UC-08	TC-008
FR-046	Multiple-course purchase	UC-08, UC-10	TC-008, TC-010
FR-047	Apply coupon codes	UC-09	TC-009
FR-048	Pricing strategies	UC-09, UC-10	TC-009, TC-010
FR-049	Free and paid support	UC-07, UC-10	TC-007, TC-010
FR-050	Stripe checkout	UC-10	TC-010
FR-051	Order history	UC-11	TC-011
FR-052	Receipts	UC-11	TC-011
FR-053	Pending order record	UC-10	TC-010
FR-054	Verified payment event	UC-43, UC-44	TC-043, TC-044
FR-055	Idempotent fulfillment	UC-43, UC-44	TC-043, TC-044
FR-056	Ask course questions	UC-19	TC-019
FR-057	Instructor answers	UC-35	TC-035
FR-058	Course announcements	UC-36, UC-20	TC-036, TC-020
FR-059	Read announcements	UC-20	TC-020
FR-060	Star ratings	UC-21	TC-021
FR-061	Written reviews	UC-21	TC-021
FR-062	Helpful review votes	UC-22	TC-022
FR-063	Completion eligibility	UC-14, UC-16, UC-45	TC-014, TC-016, TC-045

FR-to-test matrix / 4

Prior SRS FR-064 through FR-083 - designed coverage, not execution proof

FR	REQUIREMENT TOPIC	SRS USE CASES	DESIGNED TEST CASES
FR-064	Automatic PDF generation	UC-45	TC-045
FR-065	Certificate identity fields	UC-23, UC-45	TC-023, TC-045
FR-066	Unique verification code	UC-05, UC-45	TC-005, TC-045
FR-067	Public verification	UC-05	TC-005
FR-068	Certificate sharing	UC-24	TC-024
FR-069	Review submitted courses	UC-37	TC-037
FR-070	Approve/reject courses	UC-38	TC-038
FR-071	User administration	UC-39	TC-039
FR-072	Category/tag management	UC-40	TC-040
FR-073	Sales analytics	UC-41	TC-041
FR-074	Best-selling courses	UC-41	TC-041
FR-075	Completion reporting	UC-41	TC-041
FR-076	Instructor earnings	UC-34, UC-41	TC-034, TC-041
FR-077	Engagement and trends	UC-41	TC-041
FR-078	Platform settings	UC-42	TC-042
FR-079	Event-driven notifications	UC-48, UC-14	TC-048, TC-014
FR-080	Media status synchronization	UC-46	TC-046
FR-081	Payment event handling	UC-43	TC-043
FR-082	Completion issuance deduplication	UC-45	TC-045
FR-083	Instructor course statistics	UC-34	TC-034

Use-case completion register

All 48 prior UC IDs carry a named baseline test case

USE-CASE TO TEST-CASE REGISTRATION

UC-01	Register account	TC-001
UC-02	Sign in / sign out	TC-002
UC-03	Browse / filter courses	TC-003
UC-04	View course details	TC-004
UC-05	Verify a certificate	TC-005
UC-06	Update personal profile	TC-006
UC-07	Enroll in free course	TC-007
UC-08	Manage shopping cart	TC-008
UC-09	Apply coupon	TC-009
UC-10	Complete paid checkout	TC-010
UC-11	View orders / receipts	TC-011
UC-12	Access enrolled course	TC-012
UC-13	Watch video lesson	TC-013
UC-14	Record completion	TC-014
UC-15	View course progress	TC-015
UC-16	Attempt quiz	TC-016
UC-17	Take lesson notes	TC-017
UC-18	Bookmark lesson	TC-018
UC-19	Ask course question	TC-019
UC-20	Read announcements	TC-020
UC-21	Rate / review course	TC-021
UC-22	Vote review helpful	TC-022
UC-23	Download certificate	TC-023
UC-24	Share on LinkedIn	TC-024
UC-25	Create / edit course	TC-025
UC-26	Organize sections	TC-026
UC-27	Upload course media	TC-027
UC-28	Bulk upload assets	TC-028
UC-29	Set pricing / discounts	TC-029
UC-30	Preview course	TC-030
UC-31	Submit for approval	TC-031
UC-32	Maintain course content	TC-032
UC-33	Manage quizzes	TC-033
UC-34	View enrollments	TC-034
UC-35	Answer course Q&A	TC-035
UC-36	Publish announcement	TC-036
UC-37	Inspect submitted course	TC-037
UC-38	Approve / reject course	TC-038
UC-39	Manage users / roles	TC-039
UC-40	Manage categories / tags	TC-040
UC-41	View platform analytics	TC-041
UC-42	Configure settings	TC-042
UC-43	Handle payment event	TC-043
UC-44	Fulfill paid order	TC-044
UC-45	Generate certificate	TC-045
UC-46	Process media assets	TC-046
UC-47	Authorize media access	TC-047
UC-48	Dispatch notifications	TC-048

Coverage rule Every identified UC has a named TC. Additional negative tests TC-049..072 strengthen authorization, financial effects, data integrity and retry guarantees; they do not replace the positive case for a UC.

Implementation sequence & sign-off

A proposed four-week verification rollout aligned with the brief

TEST DELIVERABLES BY WEEK

WEEK	IMPLEMENTATION EMPHASIS	QUALITY OUTPUT
1	Identity, catalog, N-Tier, migrations, baseline roles	Unit fixture baseline + permission smoke + SQL migration tests.
2	Course builder, media upload/processing and moderation	Media/ownership and course-state integration suite.
3	Student lessons, pricing, Stripe, assessments, reviews	Commerce + replay integrity, quiz/progress and browser E2E.
4	Certificates, Admin analytics, Azure deployment	Award/verification, load/security/accessibility, staged UAT, release checklist.

CANDIDATE EXIT CHECKLIST

- ☐ All P0 integration, security and payment tests executed and passed.
- ☐ All 48 baseline UC-linked cases reviewed, run or formally deferred with reasons.
- ☐ No open S1/S2 defects affecting paid entitlement, authorization, moderation, quizzes or certification.
- ☐ SRS NFR-09..15 targets measured where a valid environment is provisioned; deviations risk-accepted.
- ☐ Coverage result supplied for NFR-25; SQL migrations, backup/restore and deployment smoke evidenced.
- ☐ Student, Instructor and Admin UAT representatives have signed their relevant accepted journeys.

> Evidence gate Until tests are run, this is a readiness template. The team should not present simulated pass rates, fake load charts or unapproved UAT signatures as verified system outcomes.

Open decisions & references

Review questions, source hierarchy and official engineering standards

DECISIONS TO CONFIRM BEFORE EXECUTION

DECISION	WHY IT MATTERS
Currency, coupon stacking and rounding	Defines pricing test oracles and server-side total invariants.
Quiz retake and fill-blank matching	Changes assessment cases and eligibility definition.
Certificate revocation / reissue policy	Determines one-time issuance and public verification behavior.
Approved media processing service and CDN	Determines exact media and failure test environment.
UAT participants, browsers and support level	Determines browser coverage and acceptance sampling.
CI tool/version and production-like SQL hosting	Determines test reproducibility and migration assurance.
Approval of proposed NFR numeric targets	Controls whether measured results can be labeled pass/fail.

SOURCE AND STANDARDS INDEX

[S1] EduHub project brief (attached; 5 pages)

[S2] EduHub Functional & Non-Functional Requirements v1.0

[S3] EduHub Use Case Diagram & Descriptions v1.0

[R1] Microsoft - Testing in .NET
https://learn.microsoft.com/en-us/dotnet/core/testing/

[R2] Microsoft - Code coverage with Coverlet
https://learn.microsoft.com/en-us/dotnet/core/testing/unit-testing-code-coverage

[R3] Microsoft - ASP.NET Core integration testing
https://learn.microsoft.com/en-us/aspnet/core/test/integration-tests?view=aspnetcore-10.0

[R4] Microsoft - EF Core choosing a testing strategy
https://learn.microsoft.com/en-us/ef/core/testing/choosing-a-testing-strategy

[R5] Playwright .NET - Installation and testing
https://playwright.dev/dotnet/docs/intro

[R6] Stripe - Checkout fulfillment
https://docs.stripe.com/checkout/fulfillment

[R7] OWASP ASVS 5.0.0
https://asvs.owasp.org/

[R8] W3C - WCAG 2.2 Recommendation
https://www.w3.org/TR/WCAG22/